Program overview

12-Sep-2023 12:41

Year	**2023/2024**
Organization	**Industrial Design Engineering**
Education	**Minors Industrial Design Engineering**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors Industrieel Ontwerpen	**Minors Industrial Design Engineering**						
IO-MI-075	**IO-MI-075-22 Study Abroad**						
IO-MI-076	**IO-MI-076-23 Workplacement**						
IO3820-15	Internship	30					
IO-MI-124	**IO-MI-124-23 Interactive Environment**						
IO-MI-221	**IO-MI-221-23 Advanced Prototyping**						
IO-MI-222	**IO-MI-222-23 Designing Sustainability Transitions**						
IO-MI-223	**IO-MI-223-23 Connected Creativity**						
IO-MI-240	**IO-MI-240-23 Future Mobility Design**						

Year	**2023/2024**
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Minors Industrieel Ontwerpen

Year	**2023/2024**
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IO-MI-075

Year	**2023/2024**
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Education	**Minors Industrial Design Engineering**

IO-MI-076

IO3820-15	Internship	30
Course Coordinator	J.J.A.P.M. van Domburg	
Education Period	None (Self Study)	
Exam Period	none	

Year	**2023/2024**
Organization	**Industrial Design Engineering**
Education	**Minors Industrial Design Engineering**

IO-MI-124	
Program Coordinator	Ing. A.J.C. van der Helm
Contact for students	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
Program Title	Interactive Environments Minor
Program Coordinator	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
In association with the Faculty of	BK EWI IO
EC Program	30 ECTS
Contact for Students	Aadjan van der Helm - a.j.c.vanderhelm@tudelft.nl
Exit Qualifications	The student will learn to: - formulate an ambitious yet feasible spatial design vision, project planning and strategy, accordingly plan a systematic approach to the design project characterized by the use of theories, models and coherent interpretations, while having a critical attitude, and insight into the nature of most recent science and technology - recognize and apply specific technical knowledge and skills required to complete the main assignment - integrate and employ individual, major-specific skills and knowledge in a product-oriented group work environment - develop and apply task-oriented group working skills and organization strategies, prove competence in cooperating and communicating; be able to not only adequately interact, but also have a sense of responsibility, and leadership - develop and apply skills in using images and written and spoken word in order to convey the essence of a design to others - practical skills for real-life project development - estimate impact of proposed design on culture, society and environment - deploy and user-test preliminary prototypes and the final design product - gather knowledge and insights from developing working prototypes, being both building components and industrial products - present and defend design concepts - understand the concepts of large-scale embedded networks, spatial computing and distributed interaction - evaluate minor activities and compose an essay answering a question and formulating answers to that question
Program Structure 1	The program structure consists of 5 segments. The first segment is about becoming familiar with interaction design. A field trip is organized to Ars Electronica Festival (in Linz Austria), one of Europe's significant festivals on interactive media art. In addition further research is performed into relevant topics that is presented to all minor students. The topic of the second segment is about the design and prototyping of an interactive object. The third segment is about a network of interactive objects. In the fourth segment students design and build a full scale interactive environment, Lastly the fifth segment involves fine-tuning of the interactive behaviour of the interactive environment. At the end of the minor students organize a one week exhibition presenting their final concepts to the public.
Administration by the Faculty of	IO
Minor Title	Interactive Environments
Intended for	The Minor will be offered to TU Delft Bachelor students from all faculties.
Expected prior Knowledge	Students are recommended to have interest in digital technologies as well as capability of creative thinking and solving out of the box problems.
Prerequisites Minor	Students are expected to represent a level of knowledge and skills adequately corresponding to the field of their major. All minor courses are taught in English, fluent command of spoken and written English language is required.
Minor Coherence / Goal	The main field of study in Interactive Environments Minor is multi-user interaction with smart spatial environments. The course structure is built around a group project assignment of an interactive social space, consisting of autonomously operating intelligent building components and smart products. The group project provides a framework which encompasses all course modules and individual student assignments. This delivers a balanced setup between education and practice, geared towards helping students in acquiring practical skills and knowledge needed for building, testing and evaluating experimental interactive complex spatial environments.
Minor Content 1	The minor consists of 5 courses that aim to build skills and involve design activities combined with making prototypes. In the main design course students are encouraged to formulate challenging design visions. Those visions are subsequently realized and tested in form of robust, fully operational and full-scale prototypes. Students from different faculties work together in multidisciplinary teams that design, produce, and deploy unprecedented interactive spatial installations. They receive professional guidance from specialists from involved fields and direct, hands-on access to cutting-edge technologies provided by affiliated commercial partners. In process of building the prototypes they acquire practical knowledge needed to undertake this challenge. This knowledge ranges from creative design strategies through project management skills to technical and technological means of physically constructing a complex interactive structure.
Minor content 2	This is a collaboration minor. External partners bring in real life cases and obtain IP rights to the results. At the start of the minor the staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.
Education Methods	The minor involves a mix of group work and individual work. The group work involves (roughly 26 ECTS) collaborating in research, design, programming, constructing and testing. The group work is performed in various team compositions from small teams (2 students) to large teams (10 students). The individual work (roughly 4 ECTS) involves acquiring technical skills and writing essays.
Maximum number of participants	40
Remarks Maximum number of participants	Selection takes place by means of a random draw, the following formula is used: 1/3 students of the TU Delft Faculty of Industrial Design, 1/3 other TU Delft disciplines from Constructive or Design Engineering related programs, 1/3 other disciplines. This allocation is done to meet the interdisciplinary nature of the minor . Minors of the IDE faculty are open only for students from academic programs.
Minimum number of participants	40

Year	**2023/2024**
Organization	**Industrial Design Engineering**
Education	**Minors Industrial Design Engineering**

IO-MI-221	
Responsible Program Employee	W.S. Elkhuizen
Contact for students	prototyping@tudelft.nl
EC Program	30
Program Start (Study Year)	2019
Program Goals	The Advanced Prototyping Minor brings together the most recent digital manufacturing methods and virtual/mixed reality technology, in creating functional high-fidelity prototypes. Prototypes are used in various design and engineering disciplines as an early version of a product or a system. They are built for a variety of purposes including to test an idea or concept (e.g., proof-of-concept prototypes), to represent certain aspects of an intended design (e.g., visual/ functional prototypes), and to trigger discussion in a multi-disciplinary design team (e.g., boundary objects). The course aims at equipping you with a wide range of advanced prototyping skills, techniques and materials to create prototypes, which capture both function and appearance of the intended design. To that aim, the course covers a broad interdisciplinary foundation, in-depth knowledge and practical skills, through (guest) lectures, hands-on workshops and multidisciplinary research/design projects. In addition to the knowledge and skills necessary for building prototypes, the course injects necessary critical and creative thinking to determine which of these techniques and when in the design process could be applied, by considering desirability, feasibility, viability, as well as sustainability.
Administration by the Faculty of	IO
Intended for	Advanced Prototyping Minor suits both design-oriented students with science/technology background and science/technology oriented students with art/design background. The aim is to build project groups by combining students from these different backgrounds to create an environment in which students can reflect on the contribution of their own discipline for advanced prototyping. Selection takes place by means of a draw, based on the following formula: 1/2 from Industrial Design (Engineering) 1/2 from other (engineering) disciplines This allocation is done to ensure the interdisciplinary nature of the minor. We welcome (international) exchange students. Minors of the faculty of Industrial Design Engineering are open only for students from academic programmes.
Minor Content 1	The Advanced Prototyping Minor consists of two courses: - Prototyping with/for Digital Fabrication (15EC, Q1) - Advanced Prototyping Project (15EC, Q2) Q1 In the course 'Prototyping with/for Digital Fabrication' (PDF), you will learn about advanced prototyping, in relation to 3D digitisation, generative design, digital fabrication and extended reality (VR/AR). The course includes the following 5 modules: - Low-fidelity prototyping - Digitisation - Design automation - Digital fabrication - Extended reality prototyping In the 'low-fidelity prototyping' module, you will learn different techniques for rapid prototyping (e.g. with cardboard, clay, paper, etc.), which you can apply in early design phases, and for later rapid iterations, to evaluate design ideas quickly and efficiently. In the 'digitisation' module, you will learn about and apply various 3D scanning technologies and other data capture technologies (e.g., pressure measurement) to create a digital twin of people/objects. In the 'design automation' module, you will learn parametric design (e.g., with Rhino Grasshopper), and learn about topics such as topology optimization and generative design principles. In the 'digital manufacturing' module, you will learn about and use additive and subtractive digital manufacturing technologies (e.g., 3D printing, laser cutting, CNC milling, CNC turning, waterjet cutting, etc.). Finally, in the 'extended reality prototyping' module, you will learn to use VR/AR technology at different stages of the design process: for design creation (e.g. VR sketches), for design evaluation (e.g. user testing using VR/AR), and for creating virtual and/or mixed reality applications. Q2 In the Advanced Prototyping Project course, you will apply the knowledge and skills gained from Q1 in a group project. An iterative design process will be followed, in which you will design and build (multiple) prototype(s). Besides building prototypes, you will test and evaluate the prototypes and eventually exhibit both the process and the prototypes in a public location. In Q2 you will also attend several workshops on transferable skills, such as (prototype) photography, video making and pitching. In addition, there are several workshops on material testing and user testing of prototypes. What's in it for you? - Lectures and guided workshops on various 3D scanning, parametric modeling, digital fabrication technologies, and virtual/mixed reality prototyping - Collaboration with other disciplines, including engineering, science, art and design. - Access to and use of facilities/labs, including Makerspace in the TU Delft Science Center, prototyping and modeling workshop (PMB@IDE), the Body Lab (digitization @IDE), and TU Delft VR Zone.
Minor Assessment	Both courses in the minor are assessed through either individual or group project assignment. These will be graded using a rubric. The grading of the assignments will be conducted by lecturers/tutors of the course. Benchmarking will take place in the case of multiple, parallel coaches to ensure consistency. Every individual course will have to be passed with at least 6/10.
Maximum number of participants	45

Year	**2023/2024**
Organization	**Industrial Design Engineering**
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IO-MI-222	
Minor Coordinator	Ir. C.P.J.M. Kroon
Contact for students	Ir. Caroline Kroon - c.p.j.m.kroon@tudelft.nl
Program Title	Designing Sustainability Transitions
In association with the Faculty of	IO
EC Program	30
Program Start (Study Year)	2019
Program Goals	The minor is characterised by a dual approach to sustainability transitions. The courses stimulate high-level critical thinking through lectures, talks by external experts and group discussions. On the other hand stimulating the development of practical design skills for sustainability through design courses, design masterclasses, and working in multi-disciplinary student teams on the integral design challenge. The learning goals are: - Analyse the different elements of complex sustainable systems, in the context of the challenge of addressing societal needs within our planetary boundaries. - Evaluate processes of change towards a sustainable future at different scales, from individual users to complete societal systems. - Form arguments for your own perspective on how transitions could take place. - Work with at least three different design approaches for sustainability. - Select from a wide range of state-of-the-art design tools and techniques, introduced by experts in guest lectures and workshops. - Apply different design approaches, tools and techniques in practical projects. - Gain experience in carrying out project work in multidisciplinary teams. - Apply communication, entrepreneurship, and collaboration skills towards sustainability challenges in the real world.
Program Structure 1	Throughout the semester you will work in teams composed of students from diverse educational background. A Design Project is combined with lectures, expert talks, workshops, group discussions and field trips. There is cross fertilisation of knowledge and experiences between the courses, they cannot be taken separately. This minor consists of 5 courses: - The Design Challenge (13 ECTS, Q1&2) - Sustainability Issues and Societal Change (4ECTS, Q1) - Value Sensitive Design (4ECTS, Q1) - Masterclasses (6ECTS, Q2) - Demystify Green (3ECTS, Q2)
Administration by the Faculty of	IO
Intended for	Students from Industrial Design Engineering (IDE) and other TU Delft faculties, as well as (international) students from design and engineering backgrounds, on a level comparable to TU Delft.
Minor Content 1	What would a sustainable future look like? And how can we design for it? Developing your own answers to these two questions will be at the centre of this minor. Political leaders (agreed, not all), business leaders and members of society recognize the sustainability crisis that we are in; climate change, loss of biodiversity, the threat it poses to societies, economies and our individual well-being. This minor will help you understand the way of thinking that is needed to establish change and the part you can play in responding to the challenges in front of us and ahead of us. You will learn how to use a design approach when dealing with sustainability transitions. You will learn to use specific design approaches and methods to contribute to, facilitate or lead (?) sustainability transitions. You will be stimulated to develop your own vision on sustainability (one of the so-called wicked problems) and toolkit for sustainability that fits with your own views and values. This minor will help you improve your career opportunities: international organisations see the move towards a sustainable world as a complex, multidisciplinary challenge. The minor also collaborates on exploratory projects. You will gain unique experience working on leading topics with some frontrunner organisations and larger organisations. You will learn about different concepts that are central to designing sustainability transitions. Such as transition management, complex systems, systemic design, planetary boundaries, futuring, social design, everyday (social) practices, critical materials, worldviews, values and ethics, behaviour change, participatory design and co-creation and the circular economy.
Maximum number of participants	60
Remarks Maximum number of participants	Minors of the IDE faculty are open only for students from academic programs.

Year	**2023/2024**
Organization	**Industrial Design Engineering**
Education	**Minors Industrial Design Engineering**

IO-MI-223	
Minor Coordinator	K.G. Heijne
Program Title	Connected Creativity
Program Coordinator	Katrina Heijne (K.G.Heijne@tudelft.nl)
EC Program	30
Program Start (Study Year)	2023
Prerequisites	Since the minor is a full-time programme, it cannot be combined with other courses.
Contact for Students	connectedcreativity@tudelft.nl
Program Goals	The ultimate goal of Connected Creativity is to support the development of future-ready creators to approach challenges with deliberate creativity in an interdisciplinary context. Connected Creativity is split up in two parts: Q1 Ignite and Q2 Electrified. By the end of Q1 Ignite, students will be able to: Evaluate best practices in relation to individual and teams creative path Research and compare different creativity theories and strategies. Apply creativity attitudes and skills, working both in teams and individually. Apply a deliberate creative process, methods and techniques, in all stages of a creative problem solving process in an interdisciplinary context. By the end of Q2 Electrified, students will be able to: Apply a deliberate creative process in a real, interdisciplinary context, by connecting creativity and innovation in organisations. Indicate success factors and barriers of creativity in organizations. Investigate how creativity is situated in social, political and historical contexts. Criticise different scientific perspectives on creativity.
Intended for	This minor is for those who are interested in applying deliberate creativity in an interdisciplinary context. The course asks for a certain level of curiosity which we expect all students to have, but prior knowledge on creative processes is not necessary. As we aim for an interdisciplinary approach, we hope to welcome students from different disciplines and universities.
Minor Content 1	In Q1 Ignite, the students will learn the basics of creativity through three courses. Half of the time (7.5 EC) they will spend on Deliberate Creativity (DC). This is a practical course in which the students learn about how to deliberately apply creativity in individual settings and in group settings. In the group settings, the iCPS-approach (Integrated Creative Problem Solving) will be applied through real cases from different domains and topics. The focus of this course is on creativity skills and attitudes. CreativiTree (CT) is a more knowledge-based course in which the students will learn theoretical backgrounds of creativity strategies. This course accounts for 4.5 EC. They can apply this knowledge in the aforementioned course DC. The Creative Path (CP) is a course with the focus on reflection and the development of skills from the individual to the team level. This course will be 3 EC. Students will learn theoretical models on reflection, by practicing reflection moments at different levels (individual, team, organization, culture) throughout Ignite. This course supports the other two courses in capturing knowledge, skills and attitudes and will help the students in becoming creative individuals. Q1 will start with an introduction week: a Creativity Deepdive. This week will set the stage for the entire Minor.
Minor content 2	Q2 Electrified is divided in two courses: the Practice of Creativity (PoC) and Contextualising Creativity (CC). Contextualising Creativity (CC) situates the emergence of creativity theories/practices in social, political and technological contexts. Students investigate how creativity-theories came about in the context of industrialization, expectations of artificial intelligence, and the Cold War. CC is driven by the students curiosity and ambition to learn more about the creative field. In PoC all the knowledge from Q1 will be deployed in an organizational context. The link between creativity and innovation will be made. Special focus will be on barriers, success factors and the urgency of creativity within organizations. Real organizations will be involved in the course to share real life cases. Creative Path, the course from Q1, will continue in Q2 as part of PoC. Students will keep on reflecting and continue exploring their role as future ready creator.
Minor Assessment	The grading of the assignments will be conducted by lecturers and coaches of the course. Benchmarking will take place in the case of multiple, parallel coaches and between courses to ensure consistency. All assignments will be graded using a rubric. Q2 cannot be started without active participation in Q1.
Maximum number of participants	40
Remarks Maximum number of participants	All disciplines are welcome in Connected Creativity. From engineering to psychology. From German literature to astrophysics. We aim for a division between students based on: 1/3 from the faculty of Industrial Design Engineering (TU Delft) 1/3 from Engineering disciplines (TU Delft) 1/3 other disciplines 40 students is the maximum we accept. If there are more applications we will use a random draw to place students. Minors of the IDE faculty are open only for students from academic programs.

Year	**2023/2024**
Organization	**Industrial Design Engineering**
Education	**Minors Industrial Design Engineering**

IO-MI-240	
Minor Coordinator	E.D. van Grondelle
Contact for students	automotivedesign-io@tudelft.nl
Program Title	Future Mobility Design
EC Program	30
Introduction 1	Human mobility has evolved into a complex and multi domain challenge, which cannot be addressed in the narrow scope of a single industry. This minor is the evolution of a minor Automotive Design and anticipates the path on which the industrys scope gradually enlarges from vehicle design to a holistic mobility design approach. How to design mobility means that are meaningful within society and address both individual wants and collective needs? How to design seamless mobility, while preserving the underlaying functional and emotional values of individual mobility? Fundamental questions that dictate a multidisciplinary, holistic design approach, through a context driven design process.
Introduction 2	For whom? University students from the 'designing and engineering' faculties from universities of Technology i.e., Industrial Design, Aerospace Engineering, Automotive or Mechanical Engineering and Architecture. Applications from Technology Management (TBM) are also being considered. Affinity with human mobility helps and curiosity is a must!
Introduction 3	What will you learn? The encompassing mobility domain is more complex and challenging than it ever has been, both from an internal as from an external perspective. This autonomous programme provides knowledge on automotive and mobility design and technology, and an understanding thereof in specific societal contexts. Because of the complexity of the human mobility domain and the high level of design execution in mobility-design, all knowledge, insights, and experience derived from this minor, are applicable and valuable in other complex systems and domains. You will learn how to design new meaningful products in the field of mobility. Products can be new policies and services, new vehicles in the widest possible sense, and the mobility system that incorporates all these entities. You will learn how to work with a context driven design process. Develop skills to visualize ideas appropriately throughout the design process, and to visually express all knowledge derived from that design process, through various visualization techniques. You will understand their respective roles and values in multidisciplinary design teams. Learning secondary skills like design project management and presenting your work in a professional manner, both verbally and visually. You will understand the merits and the value of design research and participate in research projects. Working in international and multidisciplinary project teams some projects, while other projects are individual. You may not always find yourself in your comfort zone.
Introduction 4	Course overview Students execute three project courses, growing in depth, width and weight, from automotive styling and technology to the design of an integrated mobility system. Those three projects together are framed by, and structured through the context driven design process Vision in Design. Together they are one full walk-through of this context driven design process. Necessary knowledge is acquired through lectures, research and skills development.
Program Structure 1	Human Mobility: The minors three project are framed by a guiding course Human Mobility, safeguarding their coherence and connection. Lectures address context driven design, design project management and program coherence. Throughout the program, students compose their own reader, their final individual deliverable. Automotive Styling: In the first project design outside-in, the emphasis is on visual qualities of a vehicle. How do people perceive and value (automotive) styling? The goal is to understand design language and the level of expertise that is required to communicate with, or fulfil a role within, the domain of vehicle design. You learn the merits of branding which you will apply on a larger scale in the next projects. Lectures include aesthetics, branding and aerodynamics. Automotive Technology: In the second design inside-out project, students design vehicles that enable people to fulfil their mobility needs in a meaningful way, in a short-term future context. Product-user interaction is investigated qualitatively as much as quantitatively. Lectures include ergonomics, safety, vehicle dynamics, technology and structures, package design and near future developments such as autonomous driving. Mobility system design: In the third Mobility Systems project, all comes together in designing a long-term future mobility system. Teams research a future context and design an integrated mobility system. After which, each team member individually designs a full vehicle within that mobility system. Ethical choices on sustainability and inclusion help the designer to take a stand. Bringing together technology, usability, and styling, elicits what the designer wants to accomplish with and for people. Lectures include the taxonomy of human goals, mobility systems (policies, services, and products), sustainability and the build environment. Automotive skills: Necessary skills, such as visualisation in the widest possible sense, are being developed in parallel in two blocks each week. One to independently (further) develop your skills, and one that addresses specifically those skills that you need in your current project.
Prerequisites Minor	Minors of the faculty of Industrial Design Engineering are open only for students from academic programmes in the third year of their Bachelor studies.
Maximum number of participants	48
Minimum number of participants	30

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